

Chapter 7.

$$7.1. (a) \log\left(\frac{S_d(n)}{S_d(n-1)}\right) = \sqrt{0.33^2 \cdot \frac{1}{365}}$$

$$= \frac{0.33}{\sqrt{365}} \approx 0.0173$$

$$(b) \log\left(\frac{S_m(n)}{S_m(n-1)}\right) = \sqrt{0.33^2 \cdot \frac{1}{12}}$$

$$= \frac{0.33}{\sqrt{12}} \approx 0.0953$$

$$7.2. t = \frac{4}{12} = \frac{1}{3}.$$

The call is exercise if and only if

$$S'(t) > K = 42. \text{ Thus}$$

$$P(S'(t) > 42) = P\left(\frac{S'(t)}{S'(0)} > \frac{42}{40}\right)$$

$$= P\left(\log\left(\frac{S'(t)}{S'(0)}\right) > \log\left(\frac{42}{40}\right)\right)$$

$$= P(Z > \log 1.05)$$

Where Z is a normal random variable with mean $\mu t = 0.12 \cdot \frac{1}{3} = 0.04$ and standard deviation $\sigma\sqrt{t} = 0.24\sqrt{\frac{1}{3}} = \frac{0.24}{\sqrt{3}}$. Thus.

$$P(S'(t) > 42) = P\left(\frac{Z - 0.04}{0.24/\sqrt{3}} > \frac{\log 1.05 - 0.04}{0.24/\sqrt{3}}\right)$$

$$\approx P(Z > 0.0634)$$

$$= 1 - \Phi(0.0634)$$

$$= 0.4747.$$

$$7.3. t = \frac{1}{3}, r = 0.08, \sigma = 0.24, K = 42, S = 40.$$

and then

$$\omega = \frac{rt + \sigma^2 t / 2 - \log\left(\frac{K}{S}\right)}{\sigma\sqrt{t}}$$

$$= \frac{0.08 \times \frac{1}{3} + 0.24^2 \times \frac{1}{3} / 2 - \log\left(\frac{42}{40}\right)}{0.24 \times \sqrt{\frac{1}{3}}}$$

$$\approx -0.0904.$$

Thus. $C \approx S\Phi(\omega) - Ke^{-rt}\Phi(\omega - \sigma\sqrt{t})$

$$= 40\Phi(-0.0904) - 42e^{-0.08/3}\Phi(-0.2289)$$

$$\approx 40 \times 0.4639 - 42 \times e^{-0.08/3} \times 0.4094$$

$$\approx 1.8137.$$

$$7.4. P = C - S + Ke^{-rt}$$

$$= S\Phi(\omega) - Ke^{-rt}\Phi(\omega - \sigma\sqrt{t}) - S + Ke^{-rt}$$

$$= S(\Phi(\omega) - 1) + Ke^{-rt}(1 - \Phi(\omega - \sigma\sqrt{t}))$$

$$t = \frac{6}{12} = \frac{1}{2}, r = 0.1, \sigma = 0.3, K = 100, S = 105.$$

then

$$\omega \approx 0.5717 \rightarrow \Phi(\omega) \approx 0.7163$$

$$\omega - \sigma\sqrt{t} \approx 0.3596 \rightarrow \Phi(\omega - \sigma\sqrt{t}) \approx 0.6404$$

Therefore

$$P = 105(0.7163 - 1) + 100e^{-0.05}(1 - 0.6404)$$

$$\approx 4.418.$$

$$7.5. (a) P\left(\frac{S(0.5)}{S'(0)} < 0.9\right) = P(Z < \log(0.9))$$

where Z is a normal random

variable with mean $\mu t = 0.06 \cdot \frac{1}{2} = 0.03$

and standard deviation $\sigma\sqrt{t} = 0.3 \cdot \sqrt{\frac{1}{2}} = 0.3/\sqrt{2}$.

thus

$$P\left(\frac{S(0.5)}{S'(0)} < 0.9\right) = P\left(\frac{Z - 0.03}{0.3/\sqrt{2}} < \frac{\log(0.9) - 0.03}{0.3/\sqrt{2}}\right)$$

$$\approx P(Z < -0.638)$$

$$= \Phi(-0.638)$$

$$= 1 - \Phi(0.638)$$

$$\approx 0.2617$$

$$(b). \text{Risk-neutral drift } r - \frac{\sigma^2}{2} = 0.05 - 0.045 = 0.005.$$

Risk-neutral volatility $\sigma = 0.3$.

$$P\left(\frac{S(0.5)}{S'(0)} < 0.9\right) = P(W < \log(0.9))$$

$$= P\left(Z < \frac{\log(0.9) - 0.005 \cdot \frac{1}{2}}{0.3 \cdot \sqrt{\frac{1}{2}}}\right)$$

$$= \Phi(-0.508)$$

$$\approx 0.3057.$$

$$A = (100 \times 0.3057 + 0 \cdot (1 - 0.3057))e^{-0.05 \times \frac{1}{2}}$$

$$= 29.99$$

7.6. $t = \frac{3}{12} = \frac{1}{4}$, $r = 0.04$, $\sigma = 0.3$, $K = 100$
 $S = 95$.

then

$$\omega = \frac{r + \sigma^2/2 + \log(K/S)}{\sigma\sqrt{t}}$$

$$= \frac{0.04 \times \frac{1}{4} + (0.3)^2 \times \frac{1}{8} + \log(\frac{100}{95})}{0.3 \cdot \sqrt{\frac{1}{4}}}$$

$$\approx -0.2003$$

thus

$$C = S\Phi(\omega) - Ke^{-rt}\Phi(\omega - \sigma\sqrt{t})$$

$$\approx 95\Phi(-0.2003) - 100e^{-0.04 \times (\frac{1}{4})}\Phi(-0.2003 - 0.3\sqrt{\frac{1}{4}})$$

$$\approx 4.0148$$

(b) Let X be a normal random variable with mean $0.05 \times \frac{1}{4} = 0.0125$ and variance $0.09 \times (\frac{1}{4}) = 0.0225$. Then

$$P(X < \log(\frac{100}{95})) = P(Z < \frac{\log(\frac{100}{95}) - 0.0125}{\sqrt{0.0225}})$$

$$= \Phi(0.2586)$$

$$\approx 0.602$$

(c) The risk-neutral drift is

$$r - \sigma^2/2 = -0.005$$

Let W be a normal random variable with mean $-0.005 \times (\frac{6}{12}) = -0.0025$ and variance $(0.3)^2 \times (\frac{6}{12}) = 0.045$. Then the no-arbitrage cost of the investment is

$$50e^{-0.04 \times 1} P(W > \log(\frac{105}{95})) P(W > 0)$$

$$= 50e^{-0.04} P(Z > \frac{\log(\frac{105}{95}) + 0.0025}{\sqrt{0.045}}) P(Z > \frac{0.0025}{\sqrt{0.045}})$$

$$= 50e^{-0.04} (1 - \Phi(0.4836))(1 - \Phi(0.0118))$$

$$\approx 7.4792$$

7.7. Let W be the random variable with mean $(r - \sigma^2/2)t = 0.06 - (0.3)^2/2 = 0.0044$ and standard deviation $\sqrt{(0.3)^2/2} = 0.3/\sqrt{2} \approx 0.2263$.

$$C = e^{-0.03} 100 P(W > \log(\frac{110}{98}))$$

$$= e^{-0.03} 100 P(Z > \frac{\log(\frac{110}{98}) - 0.0044}{0.2263})$$

$$= e^{-0.03} 100 (1 - \Phi(0.207))$$

$$\approx 40.55$$

7.8. The payoff is

$$e^{-0.03} 100 P(Z > \frac{\log(\frac{40}{38})}{0.2263})$$

$$\approx 39.82$$

7.10. (a) The risk neutral geometric Brownian motion has drift $r - \sigma^2/2 = 0.06 - 0.08 = -0.02$. Let $W = \log(\frac{S(t)}{S(0)})$ which is normal with mean -0.02 and STD 0.4 . Then no arbitrage if

$$10 = e^{-0.06} (5P(W < \log(\frac{95}{100})) + XP(W > \log(\frac{110}{100})))$$

$$10 = e^{-0.06} (5 \times 0.469 + X \times 0.386)$$

$$X = \frac{10e^{0.06} - 5 \times 0.469}{0.386} \approx 21.433$$

(b) Let X be a normal random variable with mean 0.05 and STD 0.4 .

$$P(S(t) < 95) = P(X < \log(0.95))$$

$$= P(Z < \frac{\log(0.95) - 0.05}{0.4})$$

$$= P(Z < -\frac{0.1013}{0.4})$$

$$= 1 - \Phi(0.2533)$$

$$\approx 0.4$$